

Yuqi Tan

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EDUCATION

Apr 2021-current	Stanford University, USA Postdoctoral Fellow with Dr. Garry Nolan
Aug 2015-May 2021	Johns Hopkins University, USA (Conferred date May 27, 2021) PhD with Dr. Patrick Cahan
Sept 2010-Jul 2014	The Chinese University of Hong Kong, Hong Kong SAR Bachelor of Science in Cell Molecular Biology, First Class Honors Minor: French

PERSON STATEMENT

I am a computational biologist specializing in spatial multi-omics, integrating expertise in machine learning with a deep understanding of molecular and cellular biology. My work focuses on developing scalable machine learning models and streamlined computational frameworks to tackle complex biological questions, such as improving stem cell engineering, identifying novel cell types, finding spatial biomarkers for cancer immunotherapy response, and discovering spatial mechanisms in neurological disease. I have led and will continue to lead a research team develop streamlined, scalable machine learning models and apply them to explore the spatial organization of cell types in healthy aging and diseases, leveraging spatial multi-omics and 3D pathology. Through direct mentoring nine students, designing/teaching three custom courses, and contributing to three global health projects, I remain committed to fostering a diverse, equitable, and inclusive scientific environment that empowers future researchers.

PUBLICATIONS

Google scholar link: <https://scholar.google.com/citations?user=zNOpmasAAAAJ&hl=en>

First/second author (* indicates co-first authorship, \$indicates corresponding author)

1. **Tan***, Kempchen*, Becker*, Haist, Feytaerts, Xiao, Su, Rech, Hickey, and Nolan\$. SPACEc: A Streamlined, Interactive Python Workflow for Multiplexed Image Processing and Analysis. *Nat Comm*. Aug 2025.
2. **Tan***, Wei*, Hickey, Bi, Nolan\$, and Snyder\$. Mapping the Human Intestine: A Multiplexed 3D Atlas at Single-Cell Resolution. *Working progress*.
3. Tan*, **Tan***, Herb, Wang, Kosse, Cohen, Lin, Nolan, Friedman. Unveiling the Single-Cell Multi-Omic Landscape: Investigating Leptin's Influence on Food Intake Through the Arcuate Nucleus (ARC) and Paraventricular Hypothalamic Nucleus (PVH) circuit. *Target submission in Sept 2025*
4. Zhang*, **Tan***, Hickey, Wei, Han, Synder, Nolan\$ and Ma\$. Robust AI-driven Single-cell 3D Interpolation Method. *Target submission in Sept 2025*
5. Haist*, Matusiak*, **Tan***, Zimmer, Stege, Kempchen, Mischke, Chu, Weidentahler-Barth, Barlow, Rogall, Gonzalez, Vaertsch, Goltsev, Grabbe, Hickey, and Nolan\$. Spatially organized inflammatory myeloid-CD8+ T cell aggregates linked to Merkel-cell Polyomavirus driven Reorganization of the Tumor Microenvironment. *BioRxiv. Under Review in Cancer Discovery*. Nov 2024.
6. Xiao*, Su*, **Tan***, Deng, Li, Ma, Kim, Bai, Liu, Rosoklija, Sissoko, Wu, Santiago, Dwork, Hen, Mann, Fan\$, Leong\$, Boldrini\$. Spatially resolved multi-omic profiling of human hippocampus reveals region-specific alterations in major depressive disorder. *Under Review in Nat. Neuroscience*. May 2025.
7. **Tan**, Spangler, Han, Farid, Peng, and Cahan\$. Quantitative comparison of in vitro and in vivo embryogenesis at a single cell resolution. *BioRxiv. Under Review in Development*. Feb 2024.
8. **Tan** and Cahan\$. SingleCellNet: a computational tool to classify single cell RNA-Seq data across platforms and across species. *Cell Syst*. 2019 Aug 28;9(2):207-213.e2. PMID: PMC6715530
9. Brbic*, Cao*, Hickey*, **Tan**, Snyder, Nolan\$, and Leskovec\$. Annotation of Spatially Resolved Single-cell Data with STELLAR. *Nat Methods*. Oct 2022.
10. Hickey, **Tan**, Goltsev, and Nolan\$. Strategies for Accurate Cell Type Identification of CODEX Multiplexed Imaging Data. *Front. Immunol*. Aug13 2021.
11. Kumar, **Tan**, and Cahan\$. Understanding development and stem cells using single cell-based analyses of gene expression. *Development*. 2017 144: 17- 32 PMID: PMC5278625

Co-author

1. Shaban*, Chang*, Qiu, Yeo, Song, ..., **Tan**..., Ma\$, Jiang\$, Mahmoon\$. A Foundation Model for Spatial Proteomics. arXiv. Jun 2025
2. Tan, Yin, **Tan**, Ivanov, Plucinska, Ilanges, Herb, Wang, Kosse, Cohen, Lin, Friedman. Leptin Activated Hypothalamic BNC2 Neurons Acutely Suppress Food Intake. *Nature*. 2024 Oct.
3. Han*, Zhu*, Ruan*, Zhao, **Tan**, Monte, Wei, Bi, Karathanos, Laquindanum, Charville, Lin, Ling, Ford, Nolan, Wang, Snyder. Space-map: 3D single-cell atlas reconstruction from serial sections. *Submitted to Nat Method*. May 2025.
4. Phongpreecha, Ghanem, Reiss, Oskotsky, Mataraso, De Francesco, Reincke, Espinosa, Ng, Costello, Sequoia, Razdan, Xie, Berson, Kim, Seong, Szeto, Myers, Gu, Feister, Verscaj, Rose, Sin, Oskotsky, Roger, Shu, Shome, Yang, **Tan**, Levitte, Wong, Gaudilliere, Angst, Montine, Kerner, Keller, Shaw, Sylvester, Fuerch, Chock, Gaskari, Stevenson, Sirota, Prince, Aghaeepour. AI-driven Precision Total Parenteral Nutrition with TPN2.0. *Nat Medicine*. Nov 2024
5. Tal, Gunawardana-Zeigler, Peng, **Tan**, Munoz-Perez, Offenbacher, Kastner, Ciero, Randolph, Gong, Deng, Cahan, Loeb. Inhibition of DKK-1 by WAY262611 Inhibits Osteosarcoma Metastasis. *Molecular Cancer Therapeutics*. Oct 2024
6. Offenbacher, Jackson, Hayashi, Zhang, Peng, **Tan**, Stewart, Ciero, Foley, Casero, Cahan and Loeb. Polyamine Depletion by D, L-alpha-difluoromethylornithine Inhibits Ewing Sarcoma Metastasis by Inducing Ferroptosis. *Clinical Cancer Research*. May 2025
7. Mataraso, Espinosa, Seong, Reincke, Berson, Reiss, Kim, Ghanem, Shu, James, **Tan**, Shome, Stelzer, Feyaerts, Wong, Shaw, Angst, Gaudilliere, Stevenson. A machine learning approach to leveraging electronic health records for enhanced omics analysis, *Nat Machine Intelligence*. Nov 2024
8. Mirzaei, Hiller, Stelzer, Thiele, **Tan**, Becker. Computational Approaches for Connecting Maternal Stress to Preterm Birth. *Clinics in Perinatology*. Mar 2024.
9. Yau, Hung, Wong, Huang, Wong, Chow, Tsoi, **Tan**, Li, Wing, Lai. INSIHGT: An accessible multi-scale, multi-modal 3D spatial biology platform. *Nat Comm*. Nov 2024.
10. Hickey, Haist, Horowitz, Caraccio, **Tan**, Rech, Baertsch, Rovira Clave, Zhu, Vasquez, Barlow, Agmon, Goltsev, Sunwoo, Covert, and Nolan. T Cell Mediated Curation and Restructuring of Tumor Tissue Coordinates an Effective Immune Response. *Cell Reports*. Dec 2023.
11. Haist*, Stege*, Rogall, **Tan**, Wasielewski, Klespe, Meier, Mohr, Kähler, Wichenthal, Schadenfdorf, Ugurel, Lodde, Zimmer, ..., Grabbe and Loquai. Treatment management for BRAF-mutant melanoma patients with tumor recurrence upon adjuvant therapy: A multicenter study from the prospective skin cancer registry ADOREG. *Journal for Immuno Therapy of Cancer*. 2023 Aug
12. Becker, Nassar, Espinosa, Stelzer, Fayaerts, Berson, Bidoki, Chang, Saarunya, Culos, De Francesco, Fallahzadeh, Liu, Kim, Marić, Mataraso, Payrovnaziri, Phongpreecha, Ravindra, Stanley, Shome, **Tan**, Thuraiappah, Xenochristou, Xue, Shaw, Stevenson, Angst, Gaudilliere, Aghaeepour. Large scale correlation network construction for unraveling the coordination of complex biological systems. *Nature Comp. Sci*. Apr 2023.
13. Strasser, Gibbs, Gascard, Bons, Hickey, Schürch, **Tan**, Black, Chu, Ozkan, Basisty, Sangwan, Rose, Shah, Camilleri-Broet, Fiset, Bertos, Berube, Djambazian, Li, Oikonomopoulos, Rachael Fels-Elliott, Vernovsky, Shimshoni, Collyar, Russell, Ragoussis, Stachler, Goldenring, McDonald, Ingber, Schilling, Nolan, Tlsty, Huang, and Ferri. Concerted epithelial and stromal changes during progression of Barrett's Esophagus to invasive adenocarcinoma exposed by multi-scale, multi-omics analysis. *Development Cell*. 2025.
14. Becker, Fayaerts, Stelzer, Berson, Chang, Clark, Marić, Culos, De Francesco, Espinosa, Kim, Maric, Mataraso, Payrovnaziri, Phongpreecha, Ravindra, Stanley, Shome, **Tan**, Thuraiappah, Xue, Shaw, Stevenson, Angst, Gaudilliere, Aghaeepour. CoDi: Profiling functional disruption in biological systems. *Under Review in Bioinformatics*. 2022.
15. Liu, Li, Zhang, van Sambeek, Appleton, Chiang, Ng, Khoshakhlagh, Chen, Garcia-Corral, Wu, Huang, **Tan**, Chao, Aach, Tam, Lim, Raychaudhuri, and Church. Iterative transcription factor screening enables rapid generation of microglia-like cells from human iPSC. *Nat Comm*. 2025.
16. Becker, Dai, Chang, Fayaerts, Stelzer, Zhang, Berson, Clark, De Francesco, Espinosa, Kim, Marić, Mataraso, Payrovnaziri, Phongpreecha, Ravindra, Shome, **Tan**, Thuraiappah, Xue, Mayo, Quaintance, Laborde, King, Gotlib, Wong, Angst, Shaw, Stevenson, Gaudilliere, Aghaeepour. Revealing the impact of lifestyle stressors on the risk of adverse pregnancy outcomes with multitask machine learning. *Front Pediatr*. Nov 2022.

17. Bian, Cheng, Su, **Tan**, Kim, Wang, Yoo, Blackshaw and Cahan. Signaling and transcriptional networks governing late synovial joint development *BioRxiv*. Jan 2021
18. Velazquez, LeGraw, Moghadam, **Tan**, Kilbourne, Hislop, Liu, Cats, de Sousa Lopes, Plaisier, Cahan, Kiani, and Ebrahimkhani. Gene Regulatory Network Analysis and Engineering Directs Development and Vascularization of Multilineage Human Liver Organoids. *Cell Syst*. 2020 Nov 30; S2405-4712(20)30420-8.
19. Chen, de Rocha, Cahan, Kubaczka, Hunter, Sousa, Mullin, Fujiwara, Nguyen, **Tan**, Landry, Han, Yang, Lu, Jha, Vo, Zhou, North, and Schlaeger. Transcriptome Dynamics of Hematopoietic Stem Cell Formation Revealed Using a Combinatorial Runx1 and Ly6a Reporter System. *Stem Cell Reports*. 2020 May 12;14(5):956-971. PMCID: PMC7220988
20. Liu, Cheng, **Tan**, Dong, and Bian. Ligustrazine Prevents Intervertebral Disc Degeneration via Suppression of Aberrant TGF β Activation in Nucleus Pulposus Cells. *Biomed Res Int*. 2019 Dec 2; 2019:5601734. PMCID: PMC6914881
21. Radley, Schwab, **Tan**, Kim, Lo, and Cahan. Assessment of engineered cells using CellNet and RNA-Seq. *Nat Protoc*. 2017 12:1089-1102 PMCID: PMC5765439
22. Mayer, Hu, Wang, Cardenas, **Tan**, Pan, and Bednarek, SCD1 and SCD2 Form a Complex that Functions with the Exocyst and RabE1 in Exocytosis and Cytokinesis. *Plant Cell*. 2017 Oct;29(10):2610-2625. PMCID: PMC5774579
23. Cui, Zhao, Xie, Wong, Wang, Gao, Ding, **Tan**, Ueda, Zhang, and Jiang. MON1/CCZ1-mediated Rab7 activation regulates tapetal PCD and pollen development in Arabidopsis. *Plant Physiol*. 2017 Jan;173(1):206-218 PMCID: PMC5210713
24. James, Gu, Ramirez-Vizcarrondo, Hasan, Truszkowski, **Tan**, Oupravanh, Khakhalin, and Aizenman. Valproate-induced neurodevelopmental deficits in *Xenopus laevis* tadpoles. *J Neurosci*. 2015 Feb 18;35(7):3218-29. PMCID: PMC4331635

FUNDING | SCHOLARSHIP | AWARDS

2025	NIH-Institute of Aging K99/R00 Pathway to Independence Award (Score 33; Resubmitted)
	Rising Star in Engineering Health Award
2024	Stanford Postdoctoral BioX Travel Award
2023	Stanford Cancer Institute Women's Cancer Center Innovation Award (\$50,000)
	Life Science Alliance Exchange Grant (€8,000)
	Stanford Postdoctoral Travel Award
2022	Stanford University Bio-X Travel Award
	Stanford University School of Medicine Dean's Fellowship (\$36,900)
2021	Deep Learning for Microscopy Image Analysis Workshop Scholarship
2019	Johns Hopkins Graduate Student Travel Award
2014	S.H. Ho College Eminent Scholarship 2013/14 (\$HKD 80,000)
	Science Faculty Dean's List
	Hong Kong Special Administrative Region Government Reaching Out Award (\$HKD \$10,000)
	S.H. Ho College Whole Person Development Gold Medal 13/14 (\$HKD \$12,000)
2013	Finalist for Rhodes Scholarship (Hong Kong) 2014
	S.H. Ho College Outstanding Student Scholarship (\$HKD \$10,000)
	S.H. Ho College Scholarship for Overseas Academic Activity (\$HKD \$10,000)
	Dedicated Research Exchange and Mentorship (DREAM) Program Fellowship
2012	Science Faculty Dean's List
	S.H. Ho College Outstanding Student Scholarship (\$HKD \$10,000)
	S.H. Ho College Annual Academic Scholarship (\$HKD \$10,000)
	Yasumoto International Exchange Scholarship (\$HKD \$10,000)
	Young Chi Wan Exchange Scholarship (\$HKD \$10,000)
2011	S.H. Ho College Outstanding Service Scholarship (\$HKD \$10,000)
	S.H. Ho College Outstanding Student Scholarship (\$HKD \$10,000)
2010	S.H. Ho College Matriculation Scholarships for Academic Excellence (Mainland) (\$HKD \$10,000)

CONFERENCE PRESENTATION

Oral presentation

Invited

- **Tan**, The German Cancer Research Center (Heidelberg) (2025, Bonn, Germany)
- **Tan**, University of Bonn, Cluster of Excellence ImmunoSensation Symposium (2025, Bonn, Germany)
- **Tan**, MD Anderson Data Science Hackathon (2025, Texas, USA): TLS & Where to find them
- **Tan**, Visualization and User Experience Seminars on Spatial Biology at the Harvard Medical School – the Future of 3D Tissue Maps (2025, Boston, USA): On the way to a human 3D intestine atlas
- **Tan**, MultiOmics 2024 (2024, Brisbane, Australia): *Make Computational Analysis for Multiplexed Image Easy.*
- **Tan**, EPFL (2024, EPFL, Switzerland): *Versatile and Interactive Python Workflow for 2D and 3D Multiplexed Image Analysis.*
- **Tan**, The European Molecular Biology Laboratory (2024, Heidelberg, Germany): *Best Practice for Multiplexed Image Analysis.*
- **Tan**, Yale University, Biomedical Engineering Department (2023, New Haven, CT): *A Spatial Biologist's Guide to Decode Multiplexed Image Analysis.*
- **Tan**, Columbia University, Biomedical Engineering Department (2023, New York City, NY): *A Spatial Biologist's Guide to Decode Multiplexed Image Analysis.*
- **Tan**, Cancer Grand Challenges Future Leaders Conference 2022 (2022, Barcelona, Spain): *Universal Cancer Knowledge Encyclopedia*
- **Tan**, East China Normal University (2021, Shanghai, China): *Machine-learning tools on multi-omics analysis.*
- **Tan**, Johns Hopkins University Biochemistry Cellular Molecular Biology Program Annual Retreat (2019, Cambridge, MD): *SingleCellNet: a computational tool to classify single cell RNA-Seq data across platforms and across species.*
- **Tan**, Invited talk at Cell Molecular Biology Symposium at Chinese University of Hong Kong (2018, Hong Kong, China): *A quantitative solution of cell type identity for stem cell engineering.*

Selected

- **Tan**, Single Cell Genomics Gordon Research Seminar (2024, Les Diablerets, Switzerland): *Versatile and Interactive Python Workflow for 2D and 3D Multiplexed Image Analysis.*
- **Tan**, Hong Kong Laureate Forum (2023, Hong Kong, China): *Identify structural and cellular drivers of DCIS progression.*
- **Tan**, Artificial Intelligence in Medicine (2022, Stralsund, Germany): *Annotation of Spatially Resolved Single-cell Data with STELLAR.*
- **Tan** and Cahan. International Society for Stem Cell Research Annual Meeting (2019, Los Angeles, CA): *SingleCellNet: a computational tool to classify single cell RNA-Seq data across platforms and across species.*

Poster presentation

- **Tan**, Hickey, Horowitz, and Nolan, Single Cell Genomics Gordon Conference (2022, Switzerland)
- **Tan** and Cahan. International Society for Stem Cell Research Annual Meeting (2020, Virtual)
- **Tan** and Cahan. International Society for Stem Cell Research Annual Meeting (2017, Boston)

TEACHING & MENTORING

Direct mentorship

High school	Naomi Hu @ Palo Alto Gunn High School
Undergrad	Jerry Zhu @ University of Michigan
	Zarif Azher @ California Institute of Technology
	Zahir Choudhry @ Occidental College (Current position: UG at Occidental College)
	Michael Farid @JHU (Current position: MD-PhD at JHU, MD PhD program)
	Rachel Glyzer @UChicago (Current position: PhD at Stanford, NSF scholar)
	April Zhang @JHU (Current position: continuing UG at JHU)
	Sam Cowl @KansasU (Current position: PhD at University of Virginia, NSF scholar)

Master	Yuqing Han @TsinghuaU (Current position: PhD at Rice University)
PhD	Tim Kempchen @UofHeidelberg (Current position: PhD at University of Bonn)
	Dan Peng@JHU (Current position: PhD at JHU)

Classroom teaching

	<u>Instructor</u>
Nov 2022	Single-cell Spatial-omics: with Applications to Stem Cell Engineering and Cancer Immunotherapy @Stanford Develop a course for graduate students and postdocs for spatial biology
Jan 2020	Neuroscience: A single cell story @JHU Developed a syllabus and taught for an undergraduate class on scRNA-seq analysis
Jul 2019	Biomedical Engineering: Topics in Stem Cell Biology @JHU Developed a syllabus and taught stem cell biology, developmental biology and computational tools to 27 high school students
	<u>Teaching assistant</u>
2018	Computational Stem Cell Biology @JHU Assisted in developing and teaching of computational stem cell biology to 20 undergraduate and graduate students
2017-2019	Computational Biology and Bioinformatics @JHU Assisted 90 graduate students to master basic high throughput analysis tools
2018-2019	Practical Genomics @JHU Assisted in a large classroom setting to teach researchers how to apply genomics technique in research
	<u>Guest lecture</u>
Nov 2024	BIOINF 590 Image Processing and Advanced Machine Learning for Cancer Bioinformatics @ University of Michigan Teach graph neural networks and CODEX spatial analysis
Dec 2023	BENG469 Single Cell Biology @Yale University Teach computational analysis and imaging-based spatial biology
2020 - 2022	Computational Stem Cell Biology @JHU Teach machine learning algorithms on cell type classification

PEDAGOGY AND LEADERSHIP TRAINING

2025	Arthur and Sandra Irving Cancer Immunology Symposium	The symposium covers aspects on how excellent science can be advanced by formulating questions, designing and executing studies to address these questions, hiring, mentoring, collaborating, leadership, raising funds, and publishing.
2024	Stanford University Preparing for Faculty Careers Course	The course explores different types of faculty roles and different institutional contexts and helps to develop a plan to thrive once you land a faculty position
2018	Johns Hopkins University Preparing Future Faculty Teaching Academy	The program introduces evidence-based inclusive teaching practices, provide opportunities for mentored teaching & educational research,

RESEARCH EXPERIENCES

Feb-Jun2024	Visiting Scholar, EMBL, Germany with Dr. Anna Kreshuk Devise a deep learning model for 3D segmentation of light microscopic images
May-Aug2020	Summer internship, Merck Research Laboratory Characterized spatial relationship between immune cells and cancer cells in renal cell carcinoma by developing and applying machine learning models

Jun-Aug 2014	Vienna BioCenter Summer research with Dr. David Keays, Austria Identified somatic mutations responsible for microcephaly from trio-exome sequencing and sequentially investigate expression of MAST1/2 during mouse embryonic brain development
Sept 2013- Jun 2014	Research assistant of Prof. Jiang Liwen, CUHK, Hong Kong Characterized Rab7's roles in the protein trafficking (fusion of autophagosome to lysosome) via confocal imaging and sequential image analysis
May-Aug 2013	Honorary fellow with Prof. Sebastian Bednarek, University of Wisconsin-Madison, USA Identified and characterized protein-protein interaction (SCD1 and RabE) using genetic, biochemical, and imaging approaches
Jan-May 2013	Research assistant of Prof. Carlos Aizenman, Brown University, USA Characterized and statistically analyzed tectum neurons morphological changes after Valproic acid treatment with Confocal Microscopy and sequential image analysis

ACADEMIC SERVICES

2024-2026	Single-cell Genomics Gordon Research Seminar <u>Co-chair</u> Design research program and screen applications
2019-	<u>External reviewers:</u> Nature Communications, Nucleic Acid Research, Cell Report, Stem Cell Reports, Biostatistics, Annals of Applied Statistics, Communication biology, Molecular Systems Biology, PLOS Computational Biology, Frontiers in Genetics, Gene, Journal of Translational Medicine

COMMUNITY SERVICES

2022-now	Students for Higher Education Opportunities and Representation in Training <u>Mentor</u> Provide mentorship to combatting socioeconomic and systemic inequalities in the medical and graduate school admissions processes
2021	Cardiovascular Institute Summer School <u>Admission board member</u> Select candidates research internships from 460 applicants
2018 - 2019	Biomedical Engineering Extramural Development in Graduate Education <u>Internship Liaison</u> Matched interdisciplinary PhD students with desired internships Hosted events for career development
2017 - 2019	BCMB Colloquium Team <u>Team Leader</u> Organized symposiums for PhD students (over 400 participants annually) to communicate their research Designed survey and analyze data to give presenters quantitative feedback